A Comparative Study on the Effects of Sound in a Nature Mindfulness Virtual Reality Application

VIVA Presentation for CMP9140M Research Project

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Overview

- **The Problem:** The Undervalued Role of Audio in VR Mindfulness
- **The Research Aim & Questions**
- **Methodology:** A Controlled, Within-Subjects Experiment
- **Key Implementation Challenge & Solution**
- **Quantitative Results:** PANAS & IPQ
- **Qualitative Results:** Participant Voices
- **Integrated Discussion & Conclusion**
- **Reflection & Future Work**

Research Gap

- **Context:** VR is a promising tool for mental well-being and mindfulness (Navarro-Haro et al., 2017 – *"Meditation experts try Virtual Reality Mindfulness"*)
- **Visual Focus:** Most research and design focuses on visuals.
- **The Gap:** The specific impact of auditory elements remains underexplored:
  - Villiers Bosman et al., 2023 – *"The effect of audio on the experience in virtual reality: a scoping review"* (Audio is understudied compared to Visuals, inconsistent and contradictory results, lack of high-quality controlled Studies)
  - Schwind et al., 2019 – **Methodological justification** (Presence is multi-dimensional; Spatial Presence/Involvement/Experienced Realism, IPQ is a robust and validated tool, and crucial recommendation: 7-point Likert scale *"provides more variance and sensitivity"*)

Strong positive	Positive	Neutral	Negative	Strong negative		
(1)	(2)	(3)	(4)	(5)		
Strongly agree	Agree	More or less agree	Undecided	More or less disagree	Disagree	Strongly disagree
(1)	(2)	(3)	(4)	(5)	(6)	(7)

- **Core Question:** In an identical VR nature environment, does the *type of audio* (diegetic vs. non-diegetic) change the user experience?

Research Aim & Questions

- **Aim:** To compare the effects of two audio types on emotional response and sense of presence in a VR mindfulness app.
- **Key Research Questions:**
  - How does audio type influence **perceived presence** (IPQ)?
  - Does one audio type lead to a stronger sense of **immersion**?
  - Are there differences in **emotional response** (PANAS)?
  - What are users' **personal preferences** and why?

Methodology - Experimental Design

- **Design:** Mixed-Methods Within-subjects, counterbalanced.
- **Participants:** N=26.
- **Conditions:**
  - **Nature Sounds (NS):** Diegetic soundscape (birds, wind, water).
  - **Mindfulness Audio (MA):** Non-diegetic guided meditation + 432Hz music ("Happiness Frequency").
- **Constant:** Single, high-fidelity VR forest environment (static viewpoint).
- **Measures:**
  - **Quantitative:** PANAS (pre/post), IPQ (post-condition).
  - **Qualitative:** Semi-structured interviews.

Positive and Negative Affect Schedule (PANAS-SF)

	Indicate the extent you have felt this way over the past week.	Very slightly or not at all	A little	Moderately	Quite a bit	Extremely
PANAS 1	Interested	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
PANAS 2	Distressed	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
PANAS 3	Excited	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
PANAS 4	Upset	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
PANAS 5	Strong	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
PANAS 6	Guilty	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5
PANAS 7	Scared	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5

How aware were you of the real world surrounding while navigating in the virtual world? (i.e. sounds, room temperature, other people, etc.)?

extremely aware -3 -2 -1 0 +1 +2 +3 not aware at all

moderately aware

How real did the virtual world seem to you?

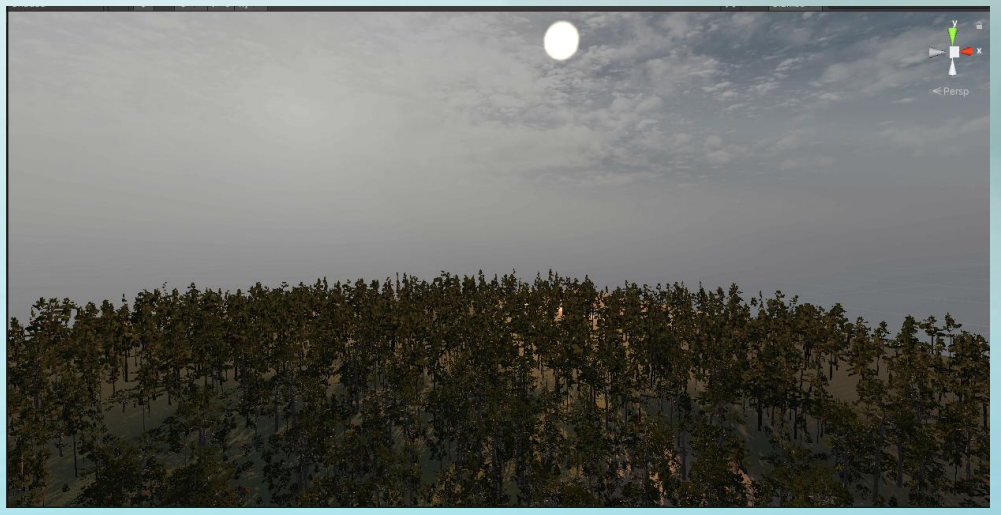
completely real -3 -2 -1 0 +1 +2 +3 not real at all

I had a sense of acting in the virtual space, rather than operating something from outside.

fully disagree -3 -2 -1 0 +1 +2 +3 fully agree

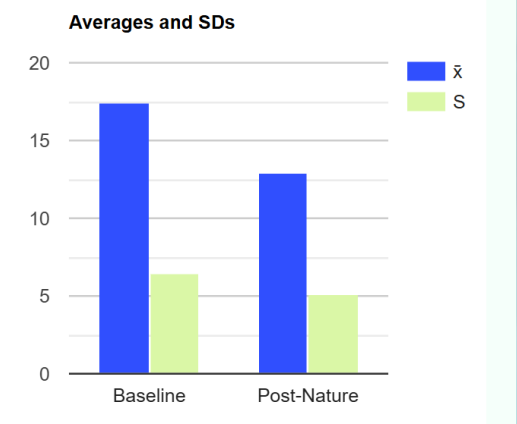
Methodology - The VR Environment

- **Initial Goal:** Build a custom, photorealistic environment from scratch.
- **The Challenge:** Severe performance issues on Meta Quest 2 (low frame rate → risk of cybersickness).
- **The Pivot:** A strategic decision to prioritise research ethics and timeline over custom artwork.
- **Final Solution:** Sourced and integrated a pre-optimised asset ('Mountain Environment Dynamic Nature') that was stable, performant, and visually appropriate.



Quantitative Results - Emotional Response (PANAS)

- **Key Finding 1:** Both conditions were **equally effective**.
- **Graph:** Bar chart showing significant reduction in Negative Affect for both NS and MA conditions (Mean Change = -4.5, $p < .01$).
- **Key Finding 2:** No significant change in Positive Affect for either condition.
- **Takeaway:** Both audio types are excellent for reducing stress and inducing calm in the short term.



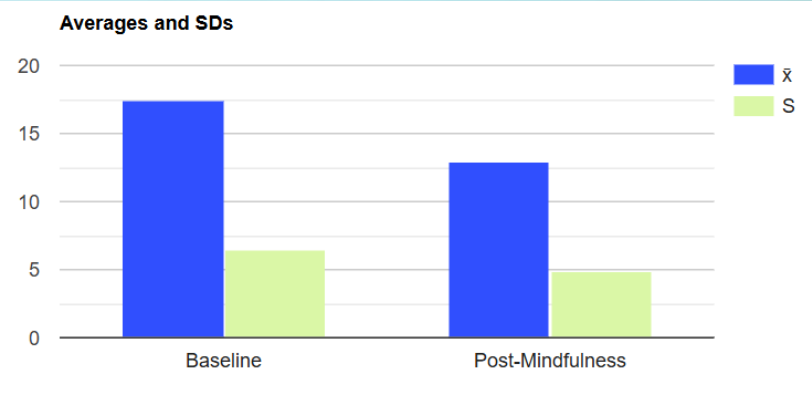


Table 5.1: PANAS Results – Emotional Response to VR Audio Conditions

Affect Type	Condition	Baseline Mean(SD)	Post-VR Mean(SD)	Mean Change	t-value	p-value	Effect Size
PA	NS	31.6 (6.1)	30.7 (6.9)	-0.9	0.70	0.497	0.135
PA	MA	31.6 (6.1)	30.5 (7.7)	-1.1	0.90	0.352	0.186
NA	NS	17.5 (6.4)	13.0 (5.2)	-4.5	3.50	0.002	0.687
NA	MA	17.5 (6.4)	13.0 (4.9)	-4.5	3.70	0.001	0.730

Quantitative Results - Sense of Presence (IPQ)

- **Key Finding:** A stark, significant divergence.
- **Table:** Side-by-side comparison of IPQ subscale scores (Spatial Presence, Involvement, Experienced Realism) showing **Nature Sounds consistently and significantly outperforming Mindfulness Audio**.
- **Highlight:** The biggest difference was in **Involvement** (NS: M=3.35 vs. MA: M=0.27, p=.002). Nature sounds created deeper psychological absorption.

Table 5.2: Mean IPQ Scores (and Standard Deviations) for Audio Conditions

Measure	Nature Sounds (NS)	Mindfulness Audio (MA)
Spatial Presence	3.69 (5.11)	1.19 (4.79)
Involvement	3.35 (4.05)	0.27 (3.67)
Experienced Realism	3.23 (4.03)	0.81 (4.28)
General Presence	2.62 (2.17)	1.08 (2.59)

Qualitative Results - Thematic Analysis (Braun and Clarke, 2006)

- **Theme 1: Immersion through Authentic Sound**
  - *Quote:* "[Nature Sounds] felt more immersive and felt more real... felt like it was all part of the same thing." (P9)
- **Theme 2: The Detaching Effect of Incongruent Audio**
  - *Quote:* "[With Mindfulness Audio] it actually felt like almost like I was sat there, but had my headphones in." (P5)
- **Theme 3: Strong Preference for Cohesion**
  - *Quote:* "The one with the nature sounds was quite a lot more immersive. It was slightly jarring in the one with the audio and no sort of sounds..." (P7)
- **Overwhelming participant preference for the Nature Sounds condition.**

Integrated Results

- **Triangulation:** The quantitative (WHAT) and qualitative (WHY) data tell one coherent story.
- **Both audio types reduce negative affect.** → Supported by participant descriptions of feeling "calm" and "peaceful".
- **Nature sounds generate significantly higher presence.** → Explained by participants feeling the sound was an "integral part" of the world, leading to a sense of "being there".
- **Mindfulness audio breaks presence.** → Explained by participants describing it as "detached" and "disconnected", a separate layer on top of the visuals.

Conclusion

- **How does audio influence presence? Dramatically.** Diegetic, environmentally congruent sound is superior.
- **Which leads to stronger immersion? Nature Sounds,** by creating a cohesive, believable world.
- **Differences in emotional response? No** for stress reduction (both good), **Yes** for the *quality* of the experience (NS more immersive).
- **User preferences? Overwhelmingly for Nature Sounds,** due to authenticity and coherence.

Reflection, Limitations & Future Work

- **Reflection:** The challenge of the implementation pivot - balancing ambition with constraints & the importance of performance in VR research.
- **Limitations:**
  - Self-report measures (potential for bias/recall).
  - Short-term exposure (no long-term data).
  - Specific environment and audio samples.
- **Future Work:**
  - Longitudinal studies.
  - Incorporate physiological measures (HRV, EDA).
  - Test different environments and audio types.
  - Explore personalisation of soundscapes.

Thank you